order25 RESULTS
PREREG sha256 5db9b84c9cabe1ac42c198d0855838bc737f091d4c354e9985c96f0a89bd7853 (hashed before numerics; RG15, RG16 hashed there).
test	value	verdict
E1 $\Gamma_\gamma^2 = I$ (exact: permutation involution, exponents sum ≡ 0 mod 9800)	true	PASS (Category A)
E2 $[\Gamma_\gamma, M] = 0$ (exact exponent multisets mod 9800)	0 mismatches of 105^2	PASS (Category A)
rank P_+ / P_-	60 / 45	—
identity $Y_- = -\frac{1}{2}D$; anti-invariance $\Gamma\, Y_- = -Y_-$	7.9e−15; 0.0	PASS
RG15 law for $Y_- = P_-$A2_corner	abs 52.031 (predicted 52.03), ‖S‖ 744, rel 0.070	**FAIL**
RG16 defect mostly anti-invariant	share_− = 0.915; Pythagoras 2.3e−15	**PASS**

Where the defect lives
‖R(A2_corner)‖ = 54.41 splits orthogonally into P_+ 15.90 and P_− 52.03: 91.5% of the squared defect is Γ-anti-invariant, i.e. sits in $\frac{1}{2}(\mathrm{Cor}_K - \mathrm{Cor}_{\{\gamma^{-1}K\}})$.
Disclosure: RG15/RG16 predictions were derived from order24 numbers (consistency checks, not blind tests).
Category A (folded into v9)
$\Gamma_\gamma^2 = I$, $[\Gamma_\gamma, M] = 0$ (exact); orthogonal split of the law residual; $Y_- = \frac{1}{2}(\mathrm{Cor}_K - \mathrm{Cor}_{\{\gamma^{-1}K\}})$.
Paper
RTA_FOUNDATION_BOUND_v9.pdf, 134 pages.

order25 NOTES
- Exact checks used exponent multisets mod 9800 (sufficient for equality); no embedding certificate needed.
- RG15 and RG16 are consistency checks of order24 values via the exact split; disclosed in PREREG.
- Near-law: not explained. The defect is mostly in the γ-anti-invariant part; next candidate (order26, to preregister): the anti-invariant combination is a difference of two cone corners that share no ray; subdivide the region between K and $\gamma^{-1}K$ (cone((7,−1),(1,0)), unimodular) and test whether the defect of Y_- is carried by that intermediate cone's corner.
- Lock respected: MAPS_CANDIDATES untouched; no Resonant objects; no bridge claim; no Residual↬Resonant candidate. E1–E3 OPEN. Ξ not identified.
- nice, OMP_NUM_THREADS=1, <3 GB, ~1 s; no push/sends; new files only.